



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

This project aimed to support SpaceY's commercial space age ambitions by developing predictive capabilities for first stage reusability and determining launch costs. Leveraging an extensive dataset collected via API and web scraping, the analysis involved comprehensive data wrangling, exploratory data analysis (EDA), interactive visualizations, and predictive modeling.

Methodology & Analysis: Data encompassing launch details, rocket types, payloads, and landing outcomes was meticulously collected and cleaned. Key activities included:

- **Data Wrangling:** Conversion of data types, handling missing values, and transforming outcomes into numerical formats.
- **Exploratory Data Analysis (EDA):** Visualizations (scatter, bar, line plots) and SQL queries explored relationships between payload mass, flight number, launch site, orbit type, year, and success rates. Feature engineering involved creating dummy variables and standardizing numerical features.
- **Interactive Visual Analytics:** Folium was used to map launch sites and landmark distances, while Plotly Dash created interactive dashboards for analyzing success rates by launch site, booster version, and payload mass.
- **Predictive Analysis:** Four classification models (Logistic Regression, SVM, Decision Tree Classifier, K-Nearest Neighbors) were trained and evaluated. Model performance was assessed using accuracy on both training and test sets, and confusion matrices were generated. Notably, the train-test split was performed with and without stratification to assess its impact on results

Executive Summary

Key Findings:

Launch Site & Landing Success:

- **CCAFS SLC 40:** Despite a period of poor performance (between Flight #20-40), it hosts the most flights and showed increasing landing success with higher flight numbers.
- **VAFB SLC 4E:** Had the fewest flights but demonstrated a good success record, with operations stopping after Flight #70.
- **KSC LC 39A:** Exhibited the highest success rate (10/13 landings), though it began operations later.
- **Proximity Analysis:** Launch sites are strategically located near coasts, with CCAFS SLC 40 being remarkably close to infrastructure (e.g., Highway: 0.20 km, Railway: 0.98 km), indicating good logistical access.

Payload & Orbit Influence:

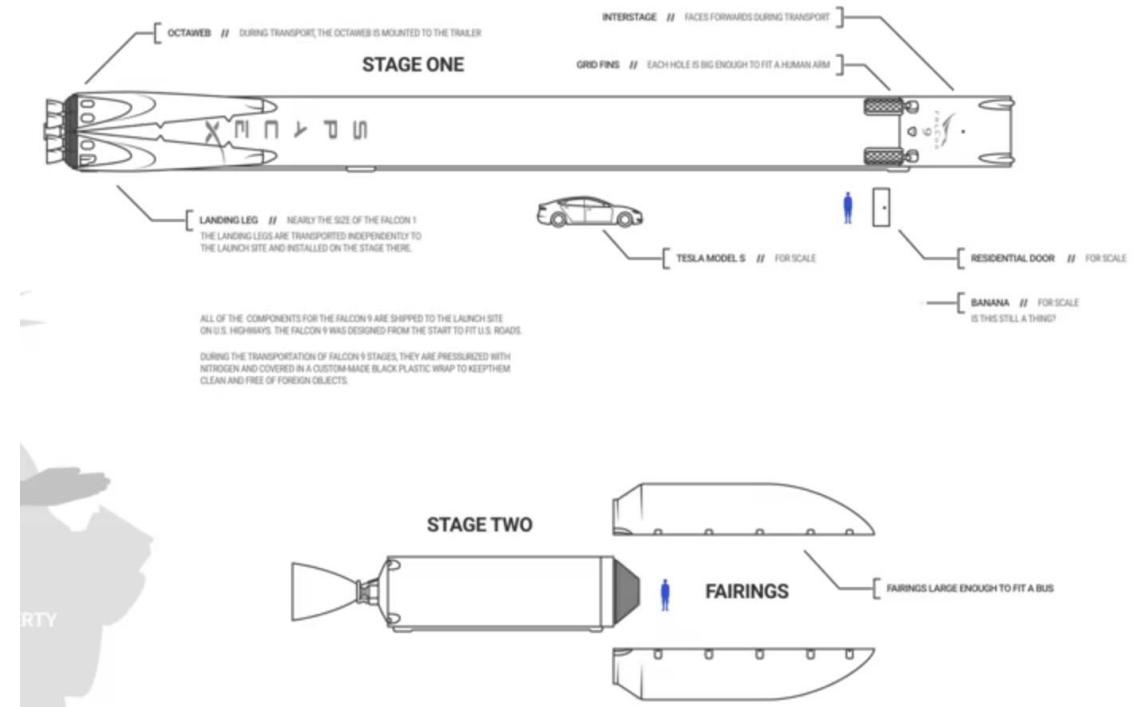
- Generally, no direct relationship was found between payload mass and landing outcome, though specific site/mass interactions were noted (e.g., KSC LC 39A not used for light rockets, VAFB SLC 4E for heavy payloads).
- **High Success Orbits:** ES-L1, GEO, HEO, and SSO orbits achieved a 100% success rate.
- **Lower Success Orbits:** GTO showed a lower success rate (around 50%).
- **LEO Orbit:** Success rate appeared to improve with the number of flights, suggesting learning from experience. For Polar, LEO, and ISS orbits, heavy payloads correlated with increased landing success.

Temporal Trends:

- Overall success rate has been consistently increasing since 2013, reaching its peak around 2020, with only a minor decrease observed in 2018.
- The best model was K-NN since it generalize better than the other models. Stratification proved to be important since without it K-NN has a test accuracy exceeding training accuracy.

Introduction

- Commercial space age is here and for our company SpaceY want to be part of it.
- In this work we will predict if we can reuse the first stage. In other words, we want to know if the first stage can land
- We will also determine the price of each launch



Section 1

Methodology

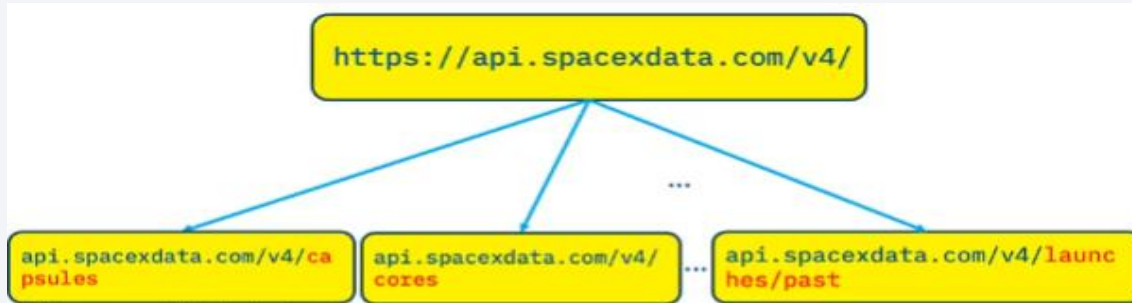
Methodology

Executive Summary

- Data collection methodology:
 - The data was collected using an API and webscraping the wikipedia page.
- Perform data wrangling
 - The data was understood by checking data types, missing values and grouping by different variables. The outcome was converted in numerical data.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - We tested 4 models and studied their accuracy in train and test sets and got the confusion matrices to determine the best model.

Data Collection

- From SpaceX REST API:



- From webscraping
wikipedia: https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

Data to be collected:

- Launches
- Rocket used
- Payload
- Launch specifications
- Landing specifications
- Landing Outcome

Data Collection – SpaceX API

- Github repository:

https://github.com/Ednamhg/DataScience_course/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Get json data using requests and convert to pandas data frame

Remove rows with multiple cores and payloads

Multiple cores or payloads

yes

Remove

no

Convert list types of cores and payloads into strings.

Convert date column into datatype and extract only the date

Restrict dates of launches to those before 11/13/2020

From other end point we change the id number to relevant data

From rocket id: booster name

From payload: mass of payload and Orbit

From launchpad: launch site name, latitude & longitude

From cores: Outcome of landing, legs, landing pad, block of the core, number of times core has been reused, serial of the core

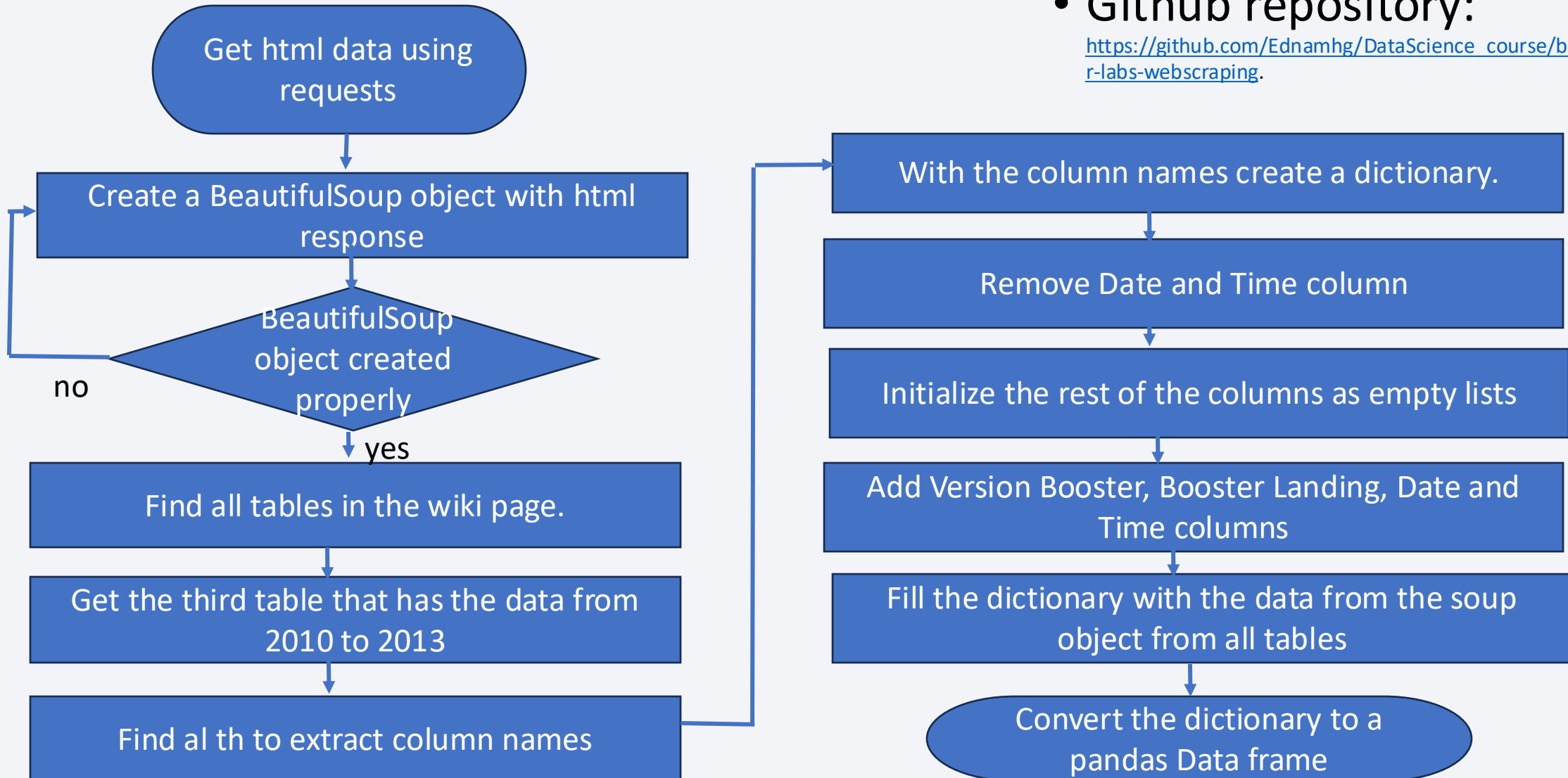
Filter data frame to include only Falcon 9 launches

Fill missing values in Payload Mass column with mean.

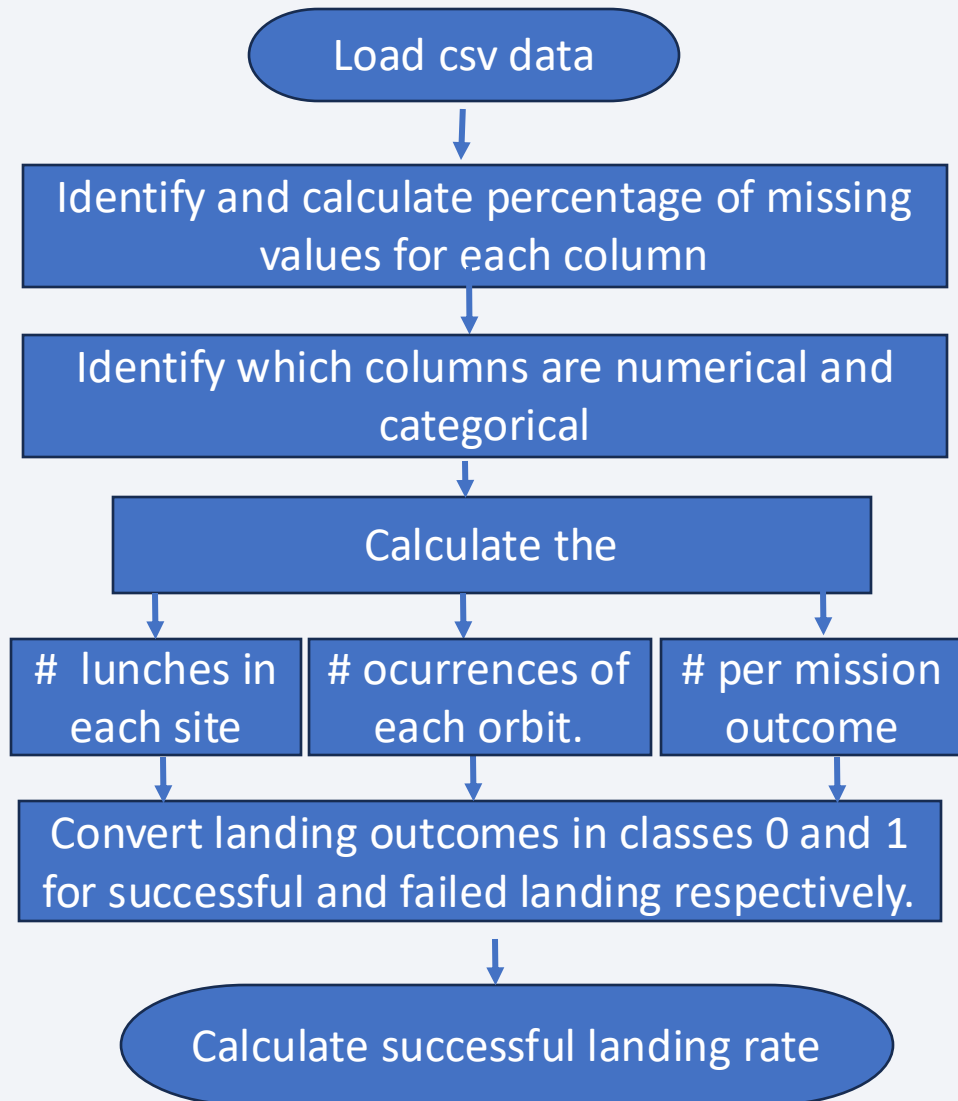
Data Collection - Scraping

- Github repository:

https://github.com/Ednamhg/DataScience_course/blob/main/jupyter-labs-webscraping.



Data Wrangling



- Github repository:

https://github.com/Ednamhg/DataScience_course/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

Important details:

- 29% of values of landing column is missing.
- Numerical: Flight Number, PayloadMass, Flights, GridFins, Reused, Legs, Block, ReusedCount, Longitude and Latitude
- Categorical: Date, Booster Version, Orbit, LaunchSite, Outcome, LandingPad, Serial
- Data in general is unbalanced
- Success rate of landing is 67 %

EDA with Data Visualization

List of Plots:

- Scatter plots with colored outcome of:
 - Pay load mass vs flight number
 - Launch site vs flight number
 - Launch site vs pay load mass
 - Orbit type vs flight number
 - Orbit type vs pay load mass
- Bar plot of success rate vs orbit type
- Line plot of Success rate vs Year

- Github repository:

https://github.com/Ednamhg/DataScience_course/blob/main/edadataviz.ipynb

Objectives:

- Learn the relationship between pay load mass, flight number, launch site, orbit type and success rate
- Since data suggested increase of success rate related to time a line plot to check this was plotted.
- Feature engineering that consists in:
 - Create dummy variables for categorical columns
 - Cast all numerical variables to float

EDA with SQL

SQL Queries :

- Names of the unique launch sites in the space mission
- 5 records where launch sites begin with the string 'CCA'
- Total payload mass carried by boosters launched by NASA (CRS)
- Average payload mass carried by booster version F9 v1.1
- Date when the first successful landing in a ground pad was achieved
- Names of the boosters which have success in drone ship and have payload mass between 4000 and 6000
- Total number of successful and failure mission outcomes
- Booster versions that have carried the maximum payload mass
- Failure landings in drone ships in 2015
- Rank landing outcomes between 2010-06-04 and 2017-03-20 in descending order

• Github repository:

https://github.com/Ednamhg/DataScience_course/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

Markers created and purpose

- Circle markers: To show launch sites in the map.
- Marker: To put a text label.
- MarkerCluster: To put the outcome of every landing for each launch location
- MousePosition: To get coordinates Latitude and longitude for a mouse over the map.
- Polyline: To mark the closest distance from a marker and a launch site

- Github repository:

https://github.com/Ednamhg/DataScience_course/blob/main/lab_jupyter_launch_site_location.ipynb

NOTE: Maps are not rendered in the Github

Build a Dashboard with Plotly Dash

Plots created using Dash:

- Pie chart for total success launches per launch site
- Pie chart of the success rate of each launch site
- Scatter plot of Success launches vs payload mass colored depending on Booster Version for all launch sites and for each launch site.

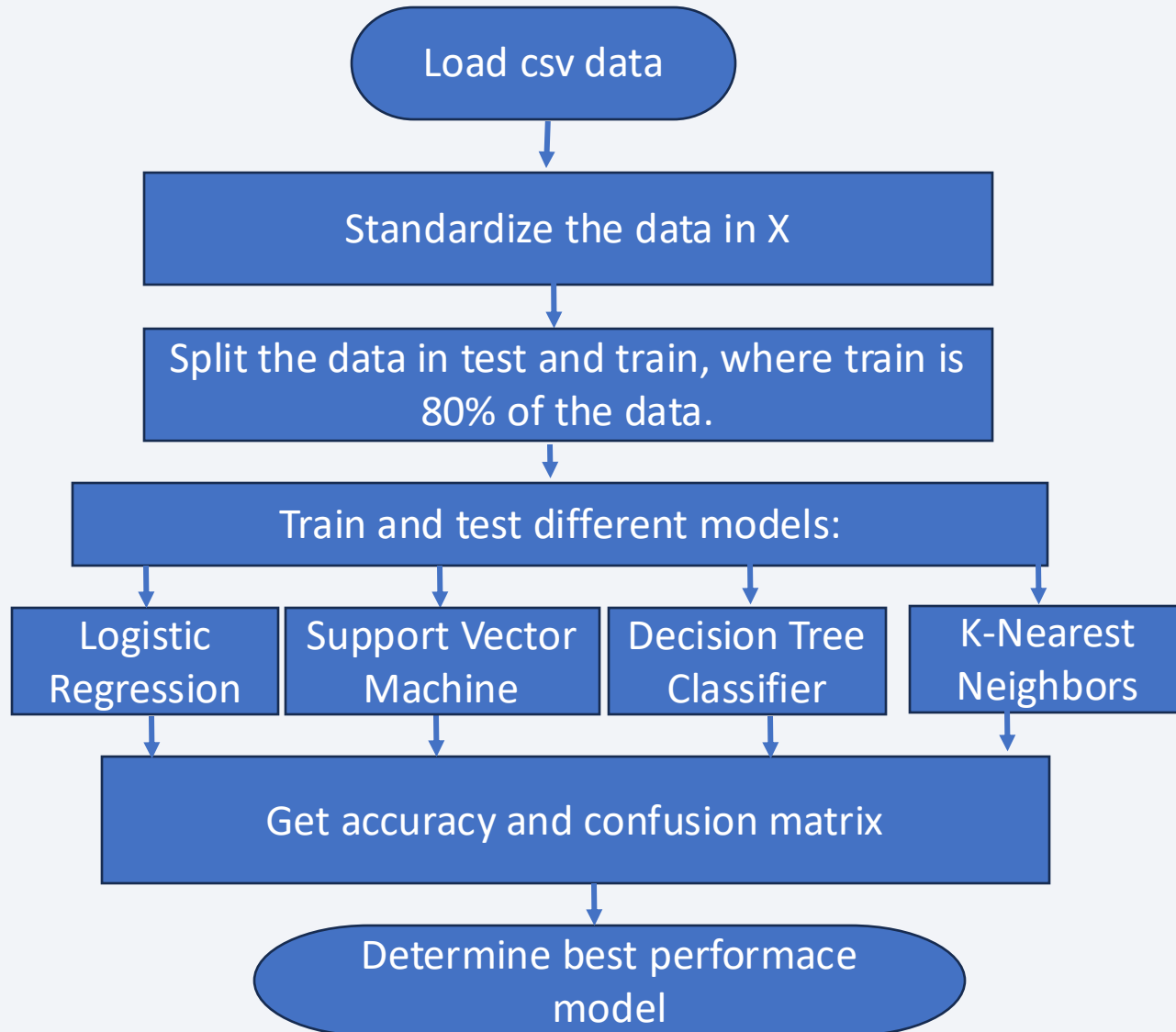
- Github repository:

https://github.com/Ednamhg/DataScience_course/blob/main/spacex-dash-app.py

Objective:

- Determine if there is any relation between the launch site and the success rate.
- Determine which booster version has more success and which payload mass has more successful rate

Predictive Analysis (Classification)



- Github repository:

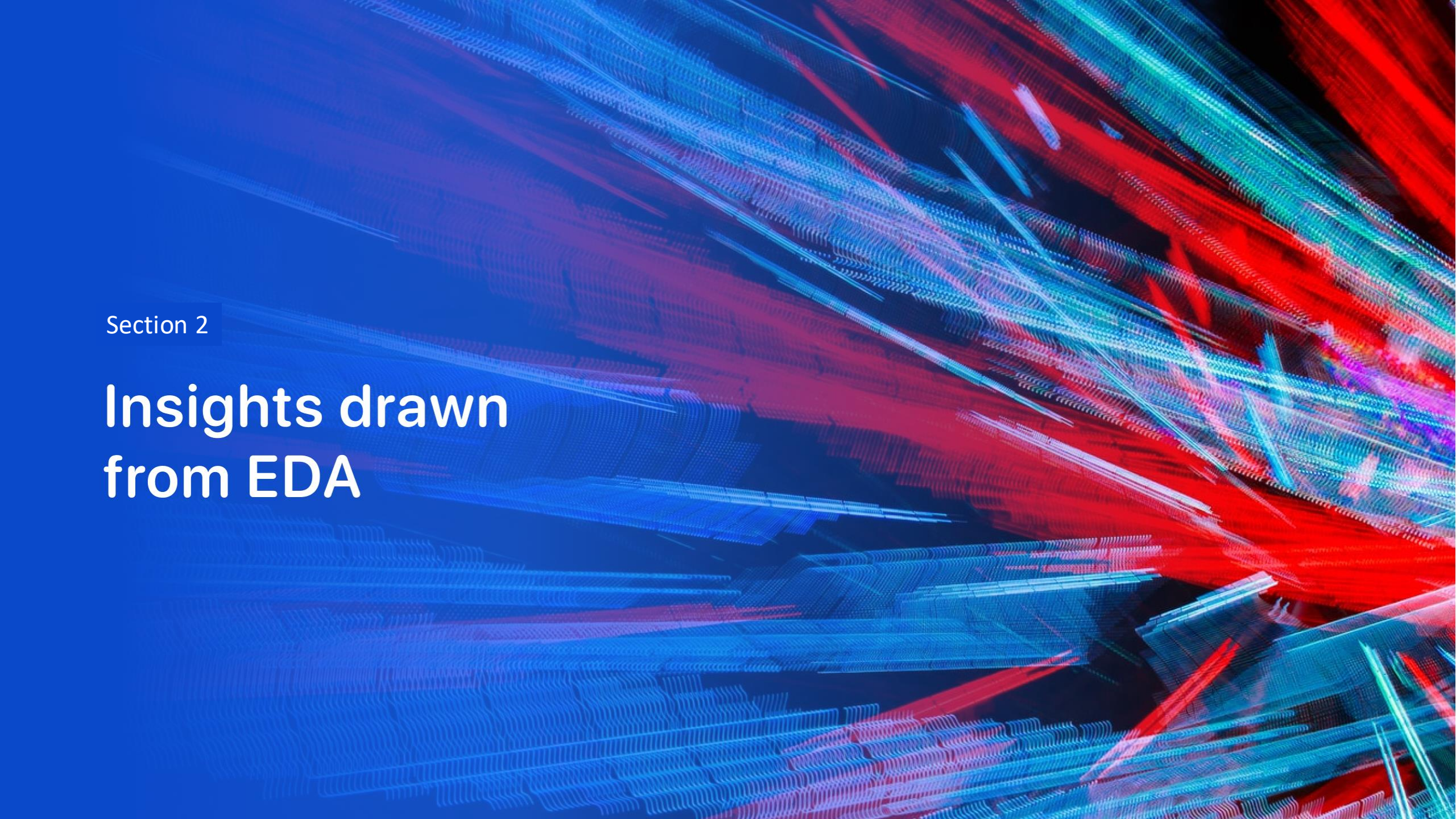
https://github.com/Ednamhg/DataScience_course/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

I got **two different results**, one where there is **no stratification** when performing the train test split and another where I **stratify the data**.

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

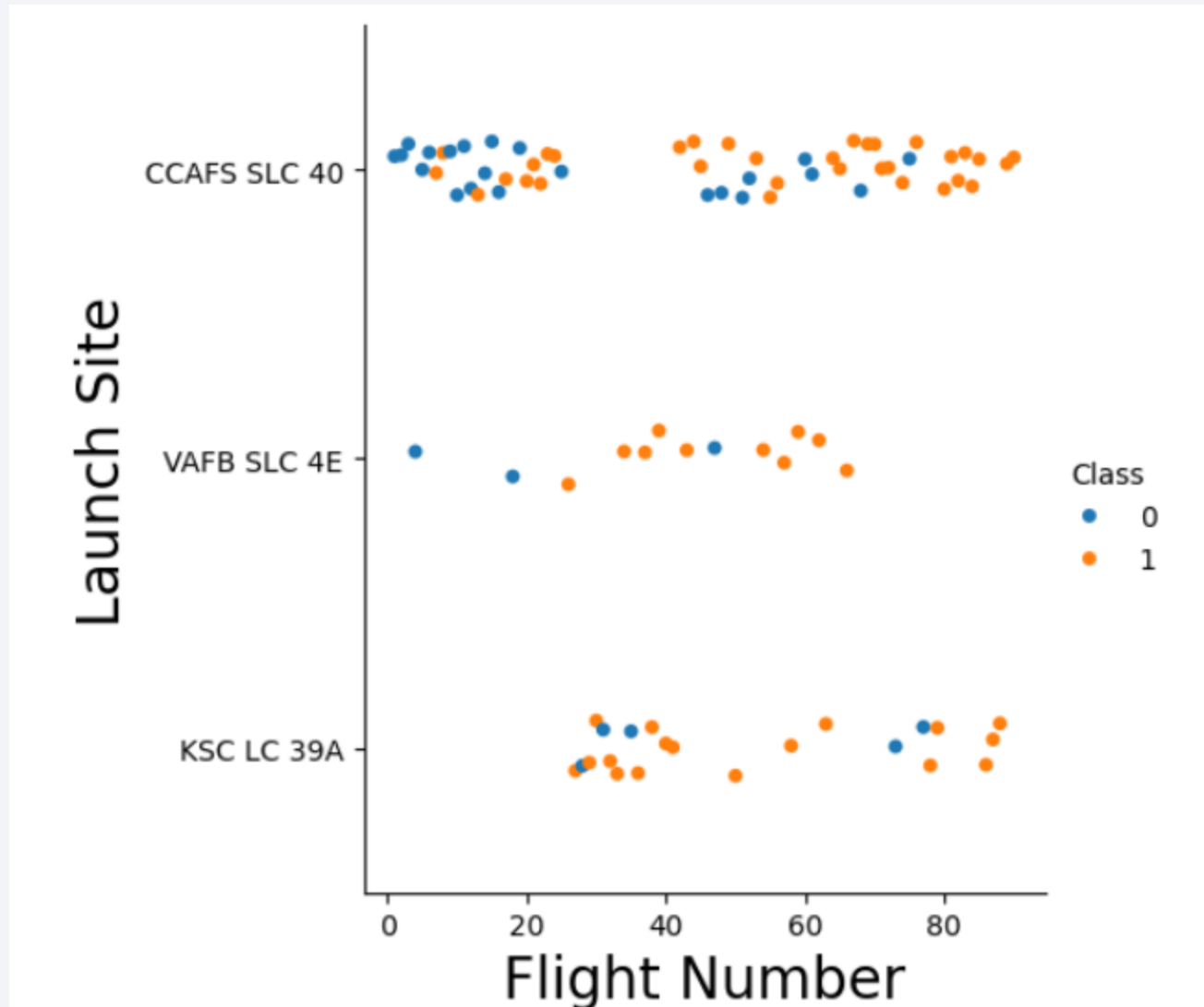
In the appendix are the results with no stratified data.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

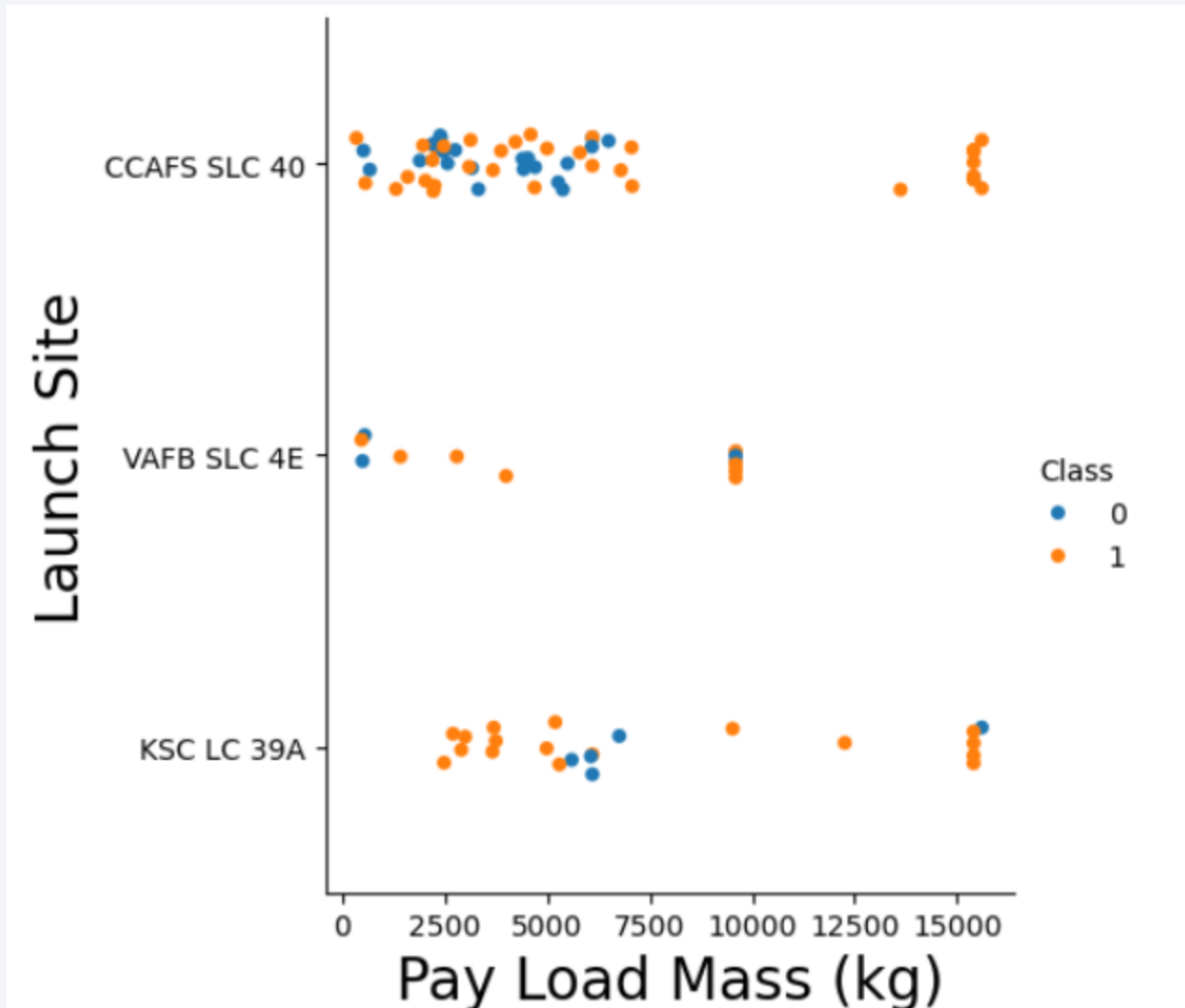
Insights drawn from EDA

Launch Site vs. Flight Number



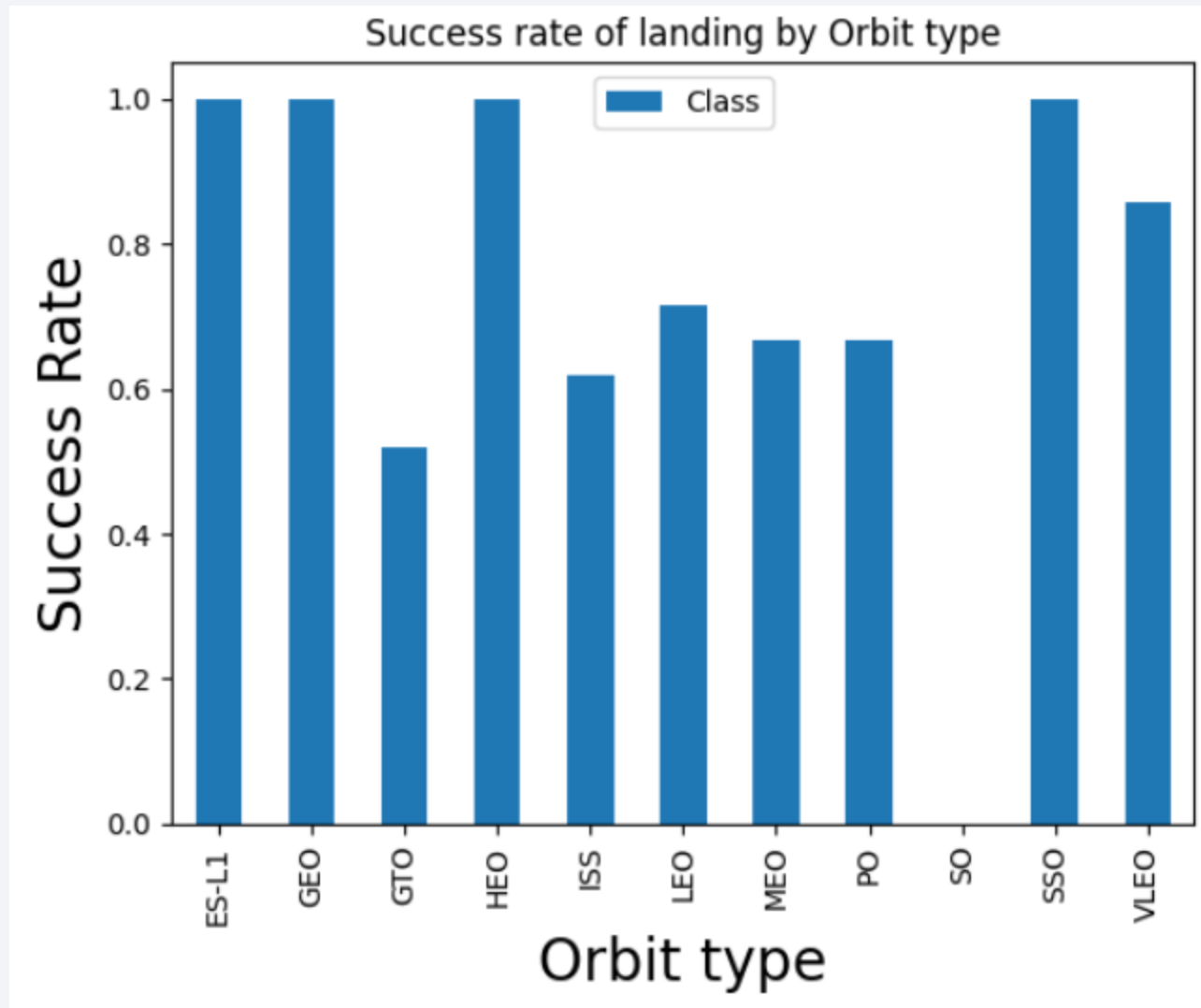
- CCAFS SLC 40 did not work for a while between Flight number 20 and 40, however it has the most number of Flights. Success landings increased in this site as number of flight increased.
- VAFB SLC 4E has the least number of Flights. It has a good record of success landings. From flight 70 flights from this launch site stopped.
- KSC LC 39A has no flight before flight number 20, The success rate is good.

Launch Site vs. Payload



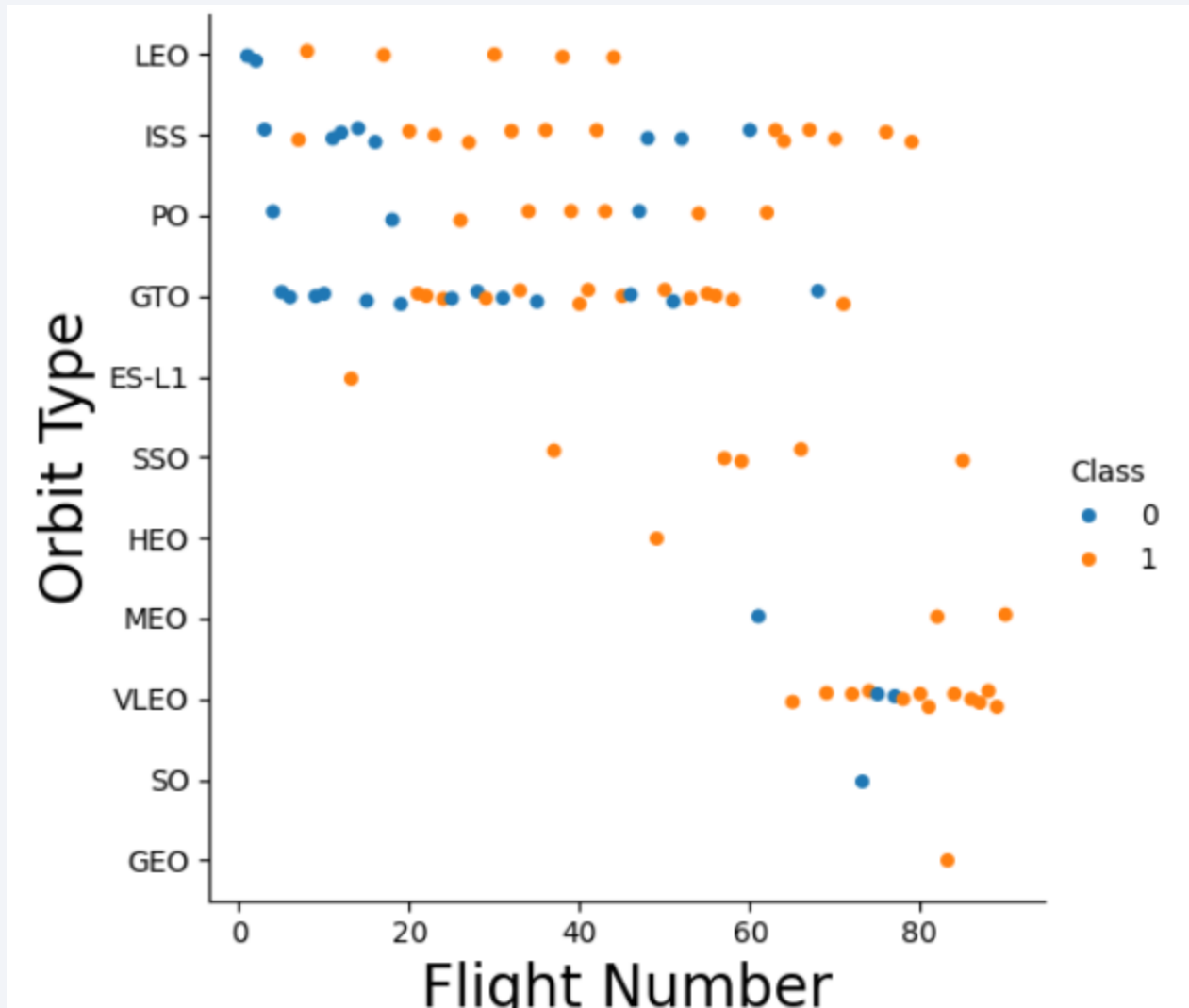
- There is no relation between payload mass and landing outcome
- CCAFS SLC 40 did not work with masses between 8 000 and 12 500 kg. The reason should be related to mission purpose, more than launch site capabilities
- VAFB SLC 4E has no heavy payload mass rockets (mass > 10 000 kg)
- KSC LC 39A does not work with light rockets (payload mass < 2000 kg) but that might be related to the time of creation of this site which is later than the other two as suggested by Payload mass vs launch site graph in the appendix.

Success Rate vs. Orbit Type



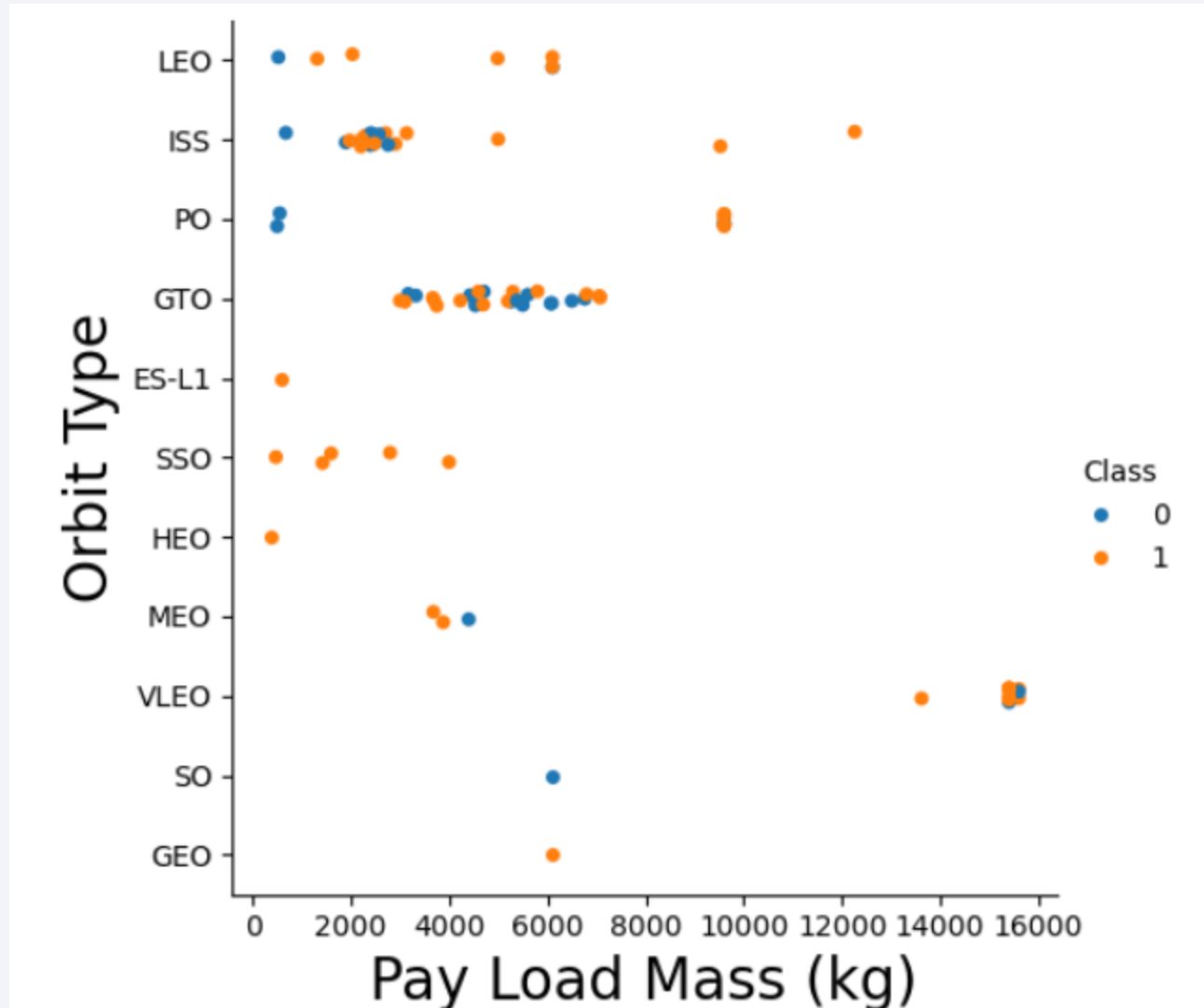
- The orbits with greater success rate are ES-L1, GEO, HEO and SSO with 100%.
- The orbit with less success rate is GTO with around 50%

Orbit Type vs. Flight Number



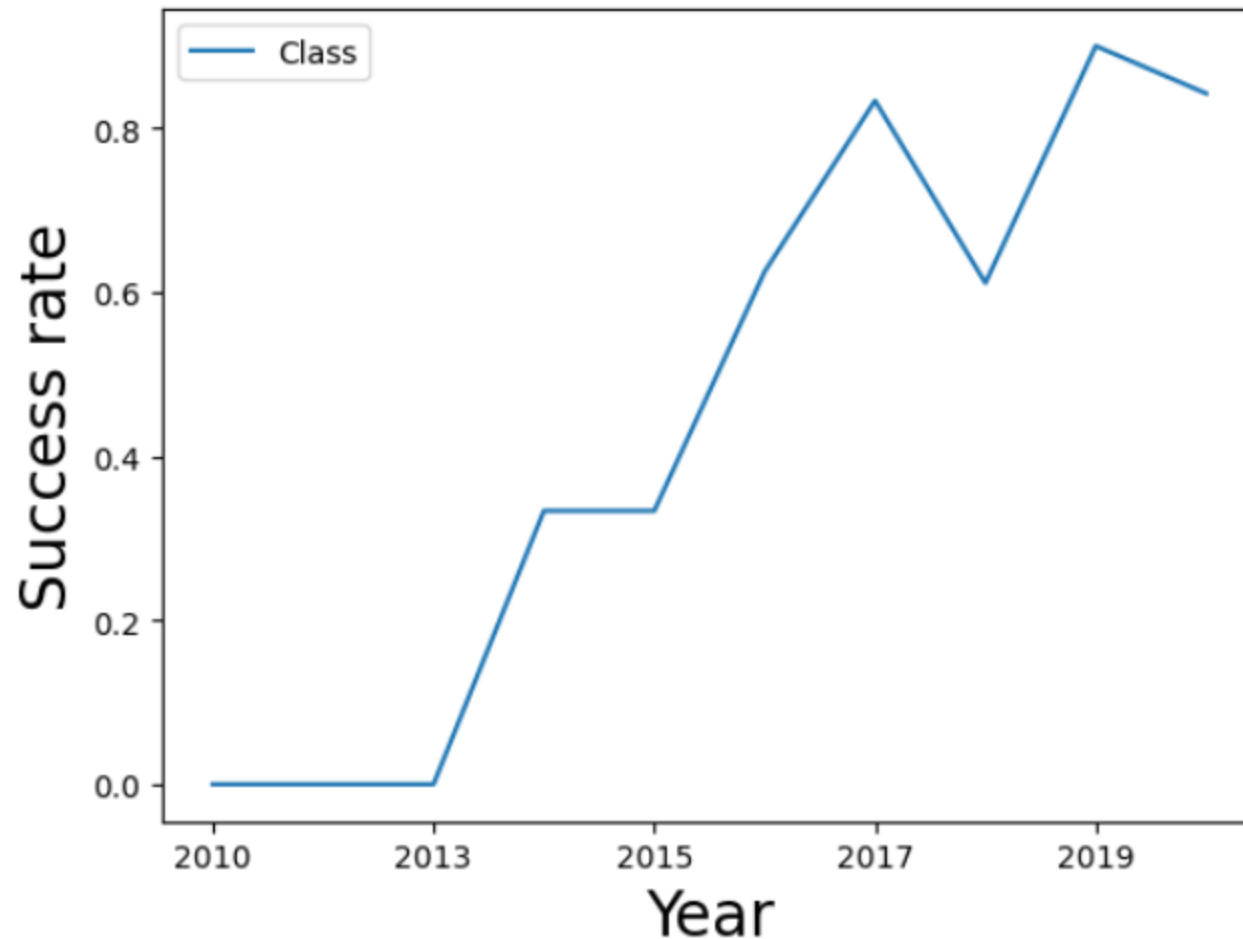
- For LEO orbit success seems to be related to the number of flights, so engeniers learned from past errors.
- No such relation is found in the other orbits.

Payload vs. Orbit Type



- With heavy payloads the successful landing increases for Polar, LEO and ISS.
- For GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.
- Heavy weights are used more for VLEO orbit
- For LEO, SSO, MEO lighter rockets are used.

Launch Success Yearly Trend



- Success rate since 2013 has been increasing till 2020.
- A small decrease of success rate is observed in 2018.

All Launch Site Names

- There are 4 launch sites only.

- **Query:** SELECT DISTINCT
"Launch_Site" FROM
SPACEXTABLE

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

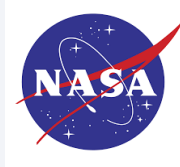
Launch Site Names Begin with 'CCA'

- The first 5 queries are from Launch site CCAFS LC-40 and LEO orbit
- The first two landing outcomes are failure and then there is no attempt.
- Since booster is not recovered the booster version is different in each case.
- Payload = 0 in the first two landings where there was an attempt to land and then is >0

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Query: SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5

Total Payload Mass

The total sum of payload mass for NASA (CRS)  is:

SUM(PAYLOAD_MASS_KG_)
45596

Query: SELECT SUM(PAYLOAD_MASS_KG_) FROM
SPACEXTABLE WHERE customer='NASA (CRS)'

Average Payload Mass by F9 v1.1

The average payload mass carried by booster version F9 v1.1 is:

Query: `SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version='F9 v1.1'`



28

AVG(PAYLOAD_MASS_KG_)

2928.4

First Successful Ground Landing Date

MIN(Date)
2015-12-22

The first successful landing outcome in ground pad was achieved in: 2015-12-22



Query: SELECT MIN(Date) FROM SPACEXTABLE WHERE
Landing_Outcome LIKE 'Success%'

Successful Drone Ship Landing with Payload between 4000 and 6000

- Names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2



Query: SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000

Total Number of Successful and Failure Mission Outcomes

Total number of successful and failure **mission** outcomes:

Successful_Outcomes	Failed_Outcomes
100	1

Query: SELECT SUM(CASE WHEN Mission_Outcome NOT LIKE 'Failure%' THEN 1 ELSE 0 END) AS Successful_Outcomes, SUM(CASE WHEN Mission_Outcome LIKE 'Failure%' THEN 1 ELSE 0 END) AS Failed_Outcomes FROM SPACEXTABLE;

Boosters Carried Maximum Payload

Names of the booster which have carried the maximum payload mass

Query: SELECT DISTINCT Booster_Version FROM SPACEXTable WHERE
PAYLOAD_MASS__KG_=(SELECT MAX(PAYLOAD_MASS__KG_) FROM
SPACEXTABLE)

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

MonthName	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Query:SELECT CASE substr(Date, 6, 2) WHEN '01' THEN 'January' WHEN '02' THEN 'February' WHEN '03' THEN 'March' WHEN '04' THEN 'April' WHEN '05' THEN 'May' WHEN '06' THEN 'June' WHEN '07' THEN 'July' WHEN '08' THEN 'August' WHEN '09' THEN 'September' WHEN '10' THEN 'October' WHEN '11' THEN 'November' WHEN '12' THEN 'December' ELSE 'Unknown Month' END AS MonthName, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTable WHERE substr(Date,0,5)='2015' AND Landing_Outcome='Failure (drone ship)'

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing outcomes count between the date 2010-06-04 and 2017-03-20, ranked in descending order

Landing_Outcome	TIMES
Precluded (drone ship)	1
Failure (parachute)	2
Uncontrolled (ocean)	2
Controlled (ocean)	3
Success (ground pad)	3
Failure (drone ship)	5
Success (drone ship)	5
No attempt	10

Query: SELECT Landing_Outcome, COUNT(Landing_Outcome) AS TIMES FROM SPACEXTable WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY TIMES

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

Location of Launch Sites

The launch sites are all in the USA near the coast either the Pacific or the Atlantic.



Success rate of launches in each launch site



The success rate of landing the first stage:

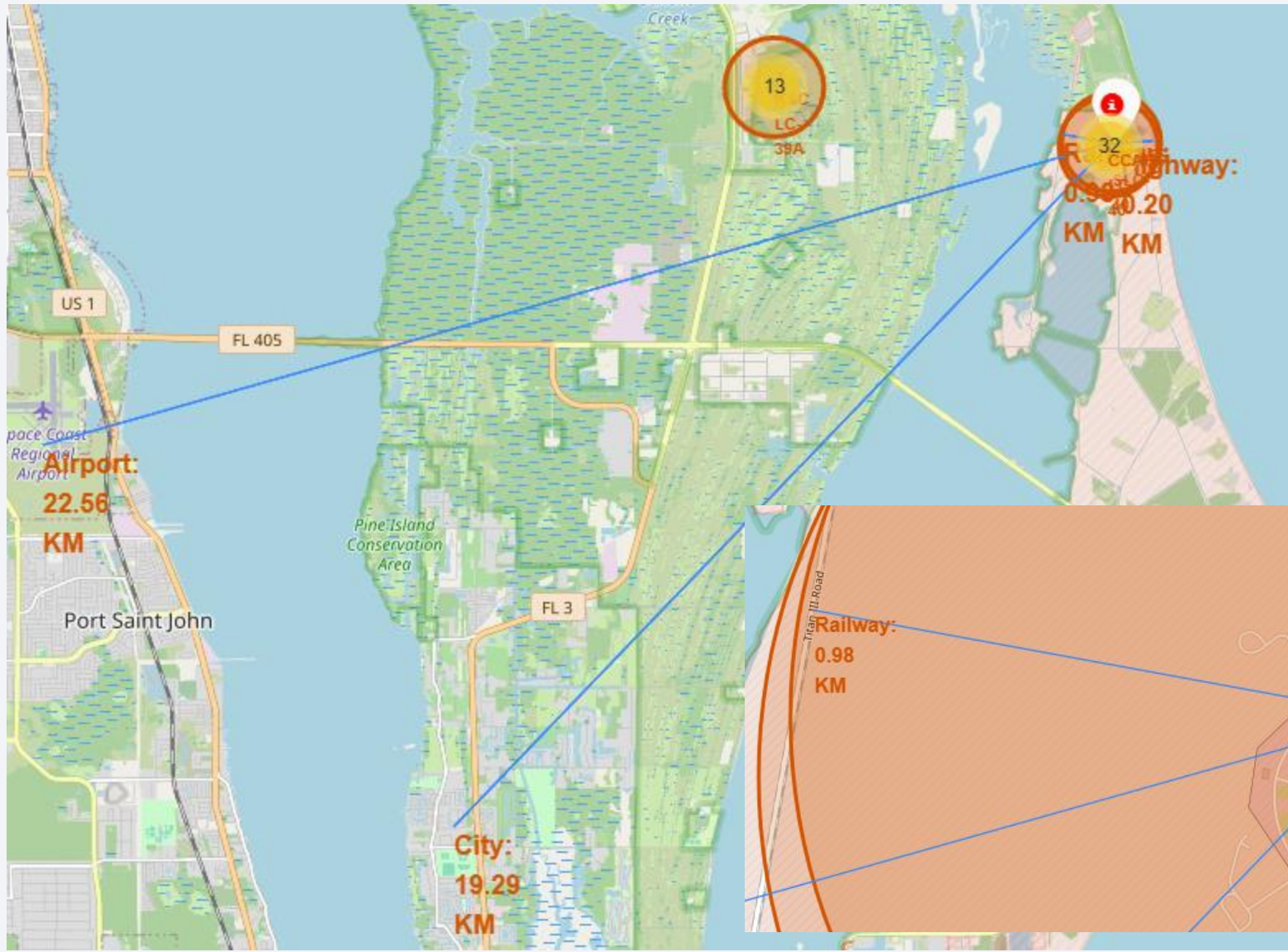
VAFV SLC is 4/10

CCAF SLC-40 is 3/7

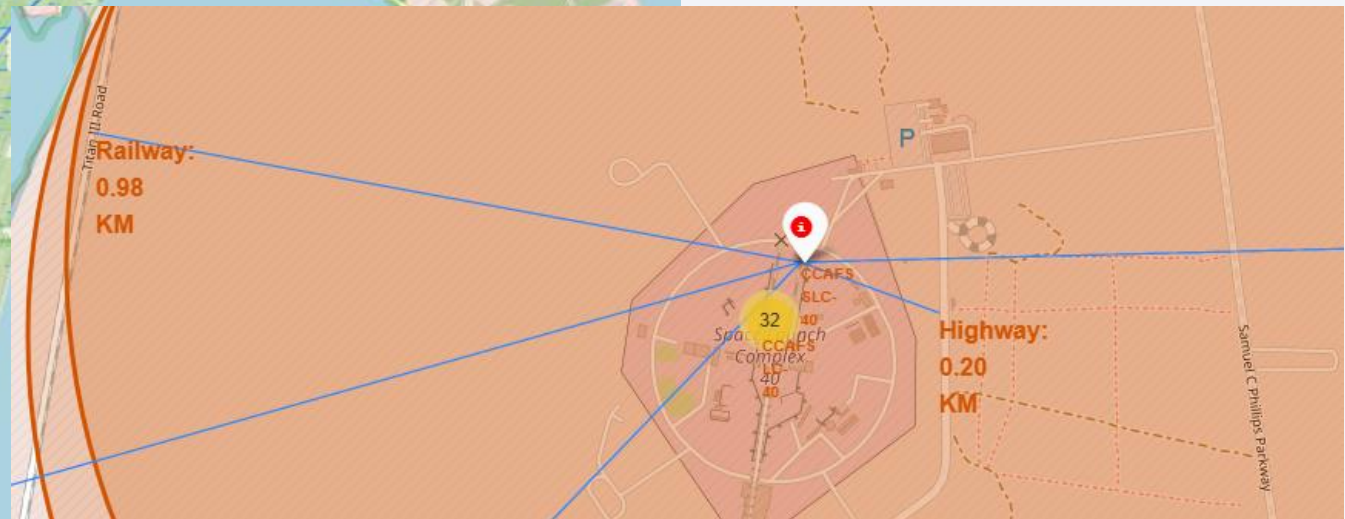
CCAF LC-40 is 7/23

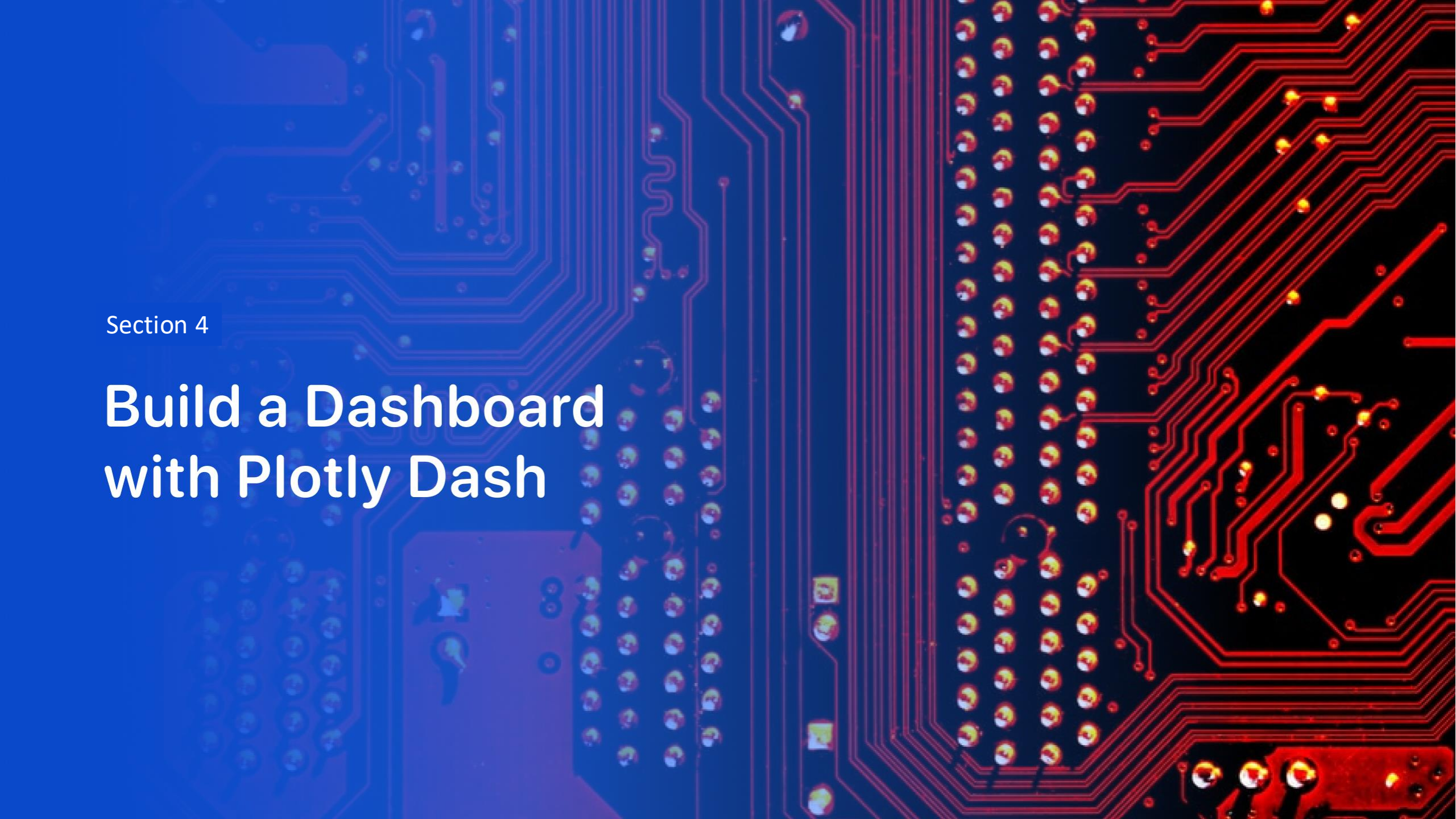
KSC LC-39A is 10/13, this site has the highest success rate

Distances to different landmarks.



Highway: 0.20 km
Airport: 22.56 km
City: 19.29 km
Railway: 0.98 km



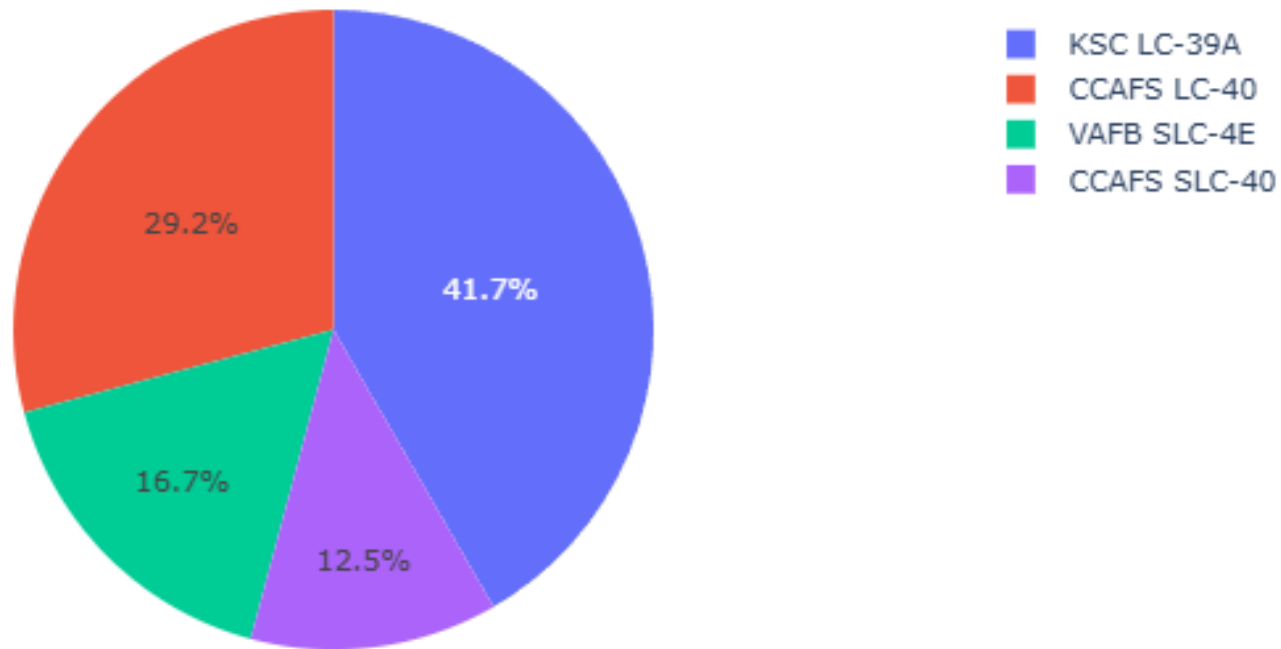


Section 4

Build a Dashboard with Plotly Dash

Success Landings per launch site

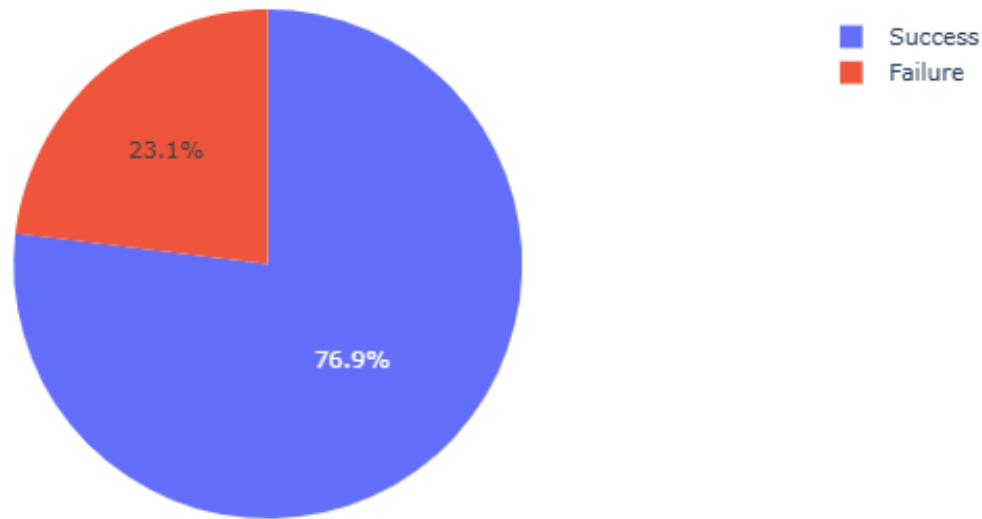
Success Landings



- Most of the rockets that present successful landings were launched from KSC. That is remarkable since CCAFS LC-40 presents more launches than KSC LC-39A
- The fewest Rockets with successful landings were launched from LC-39A CCAFS SLC-40

Success ratio for landings on launch site KSC LC-39

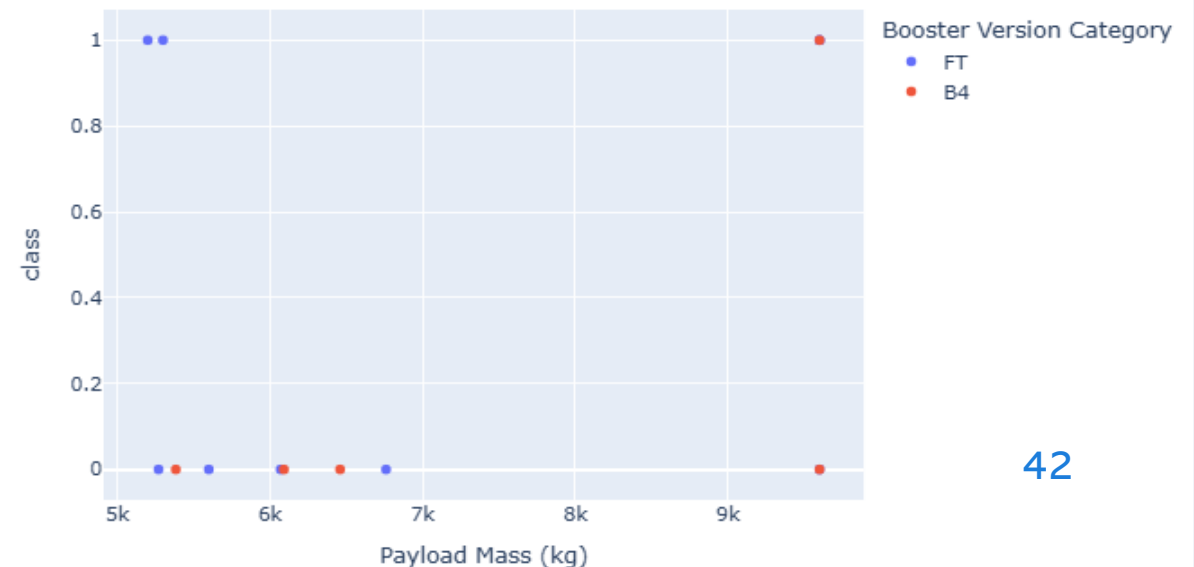
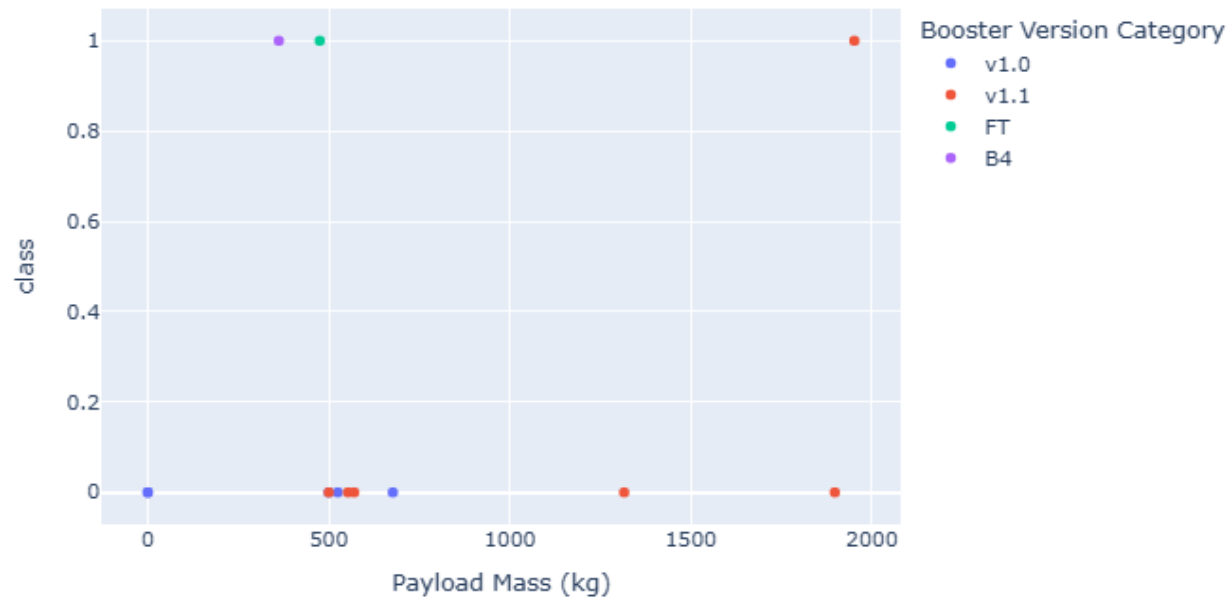
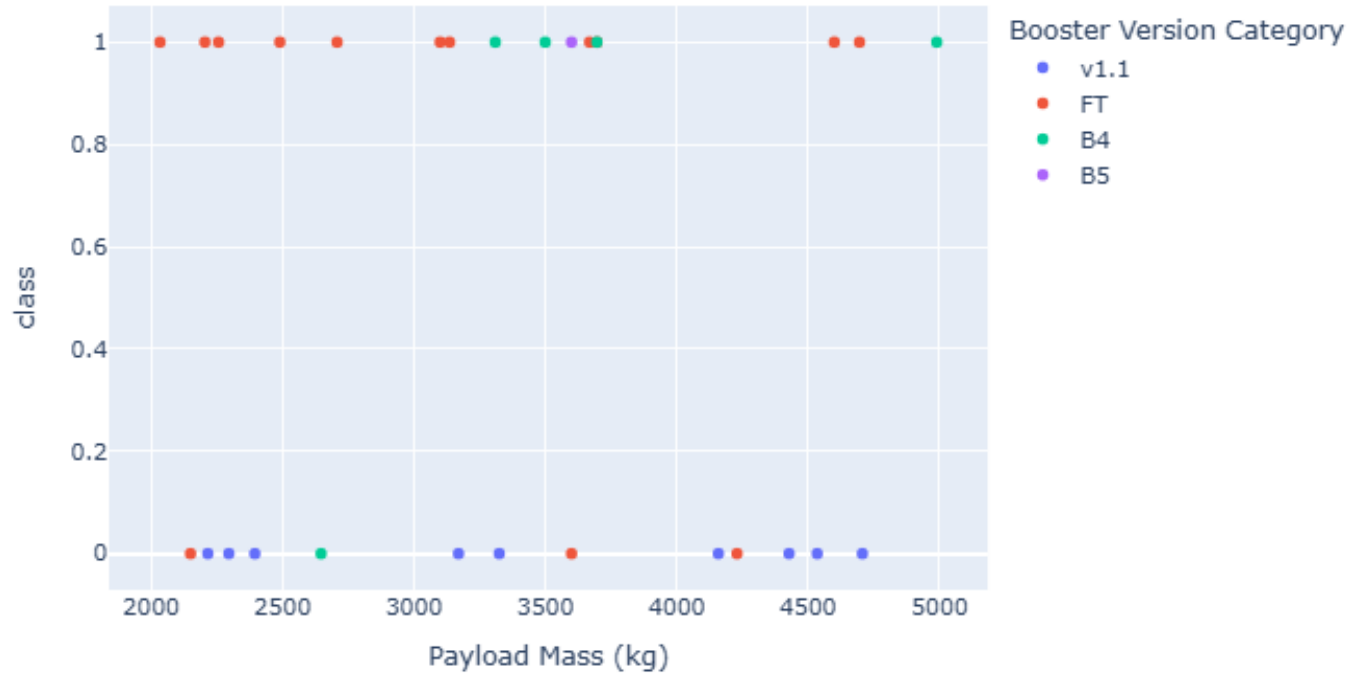
Total Successes for KSC LC-39A



- The success ratio for KSC LC-39 is more than 75%. That is impressive, but it can be related also that it started working later than the other launch sites.
- Further studies focused only in the dates the KSC LC-39 was working and landing types should be done to determine what is the reason for this success rate.

Launch Outcome vs payload for different booster versions

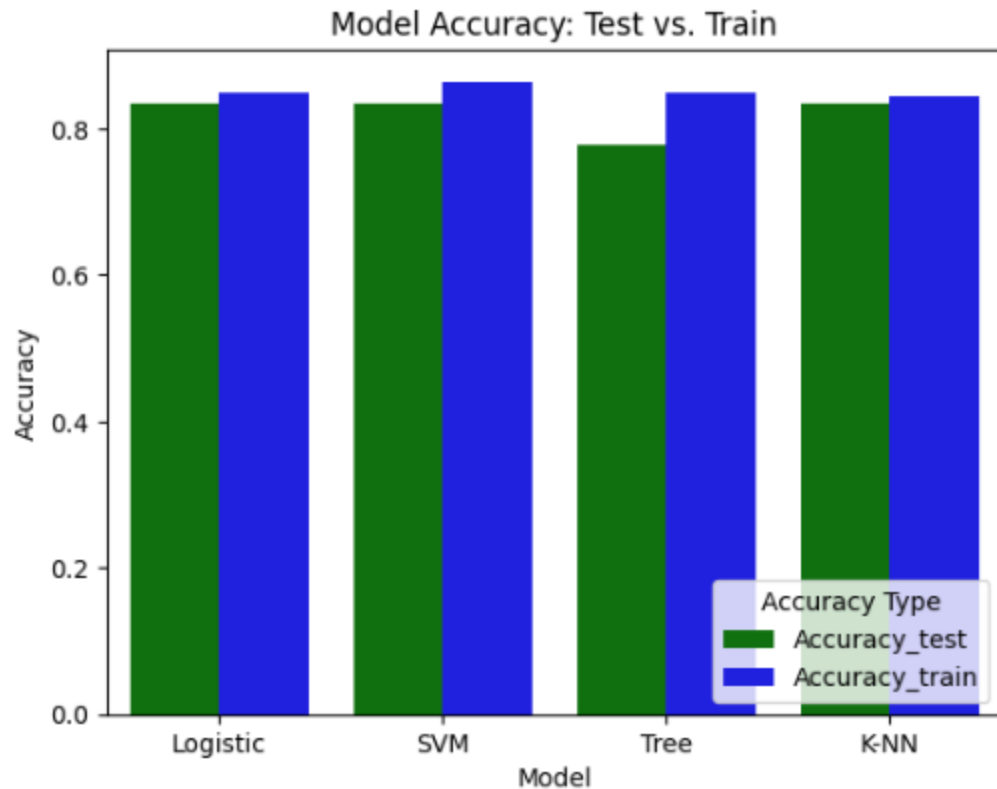
- The payload with the highest launch success rate is from 2000 to 5000 kg
- Booster FT has the highest success rate
- The payload with the lowest launch success rate are from 0 to 2000 kg and from 5000 to 10000 kg



Section 5

Predictive Analysis (Classification)

Classification Accuracy using stratify

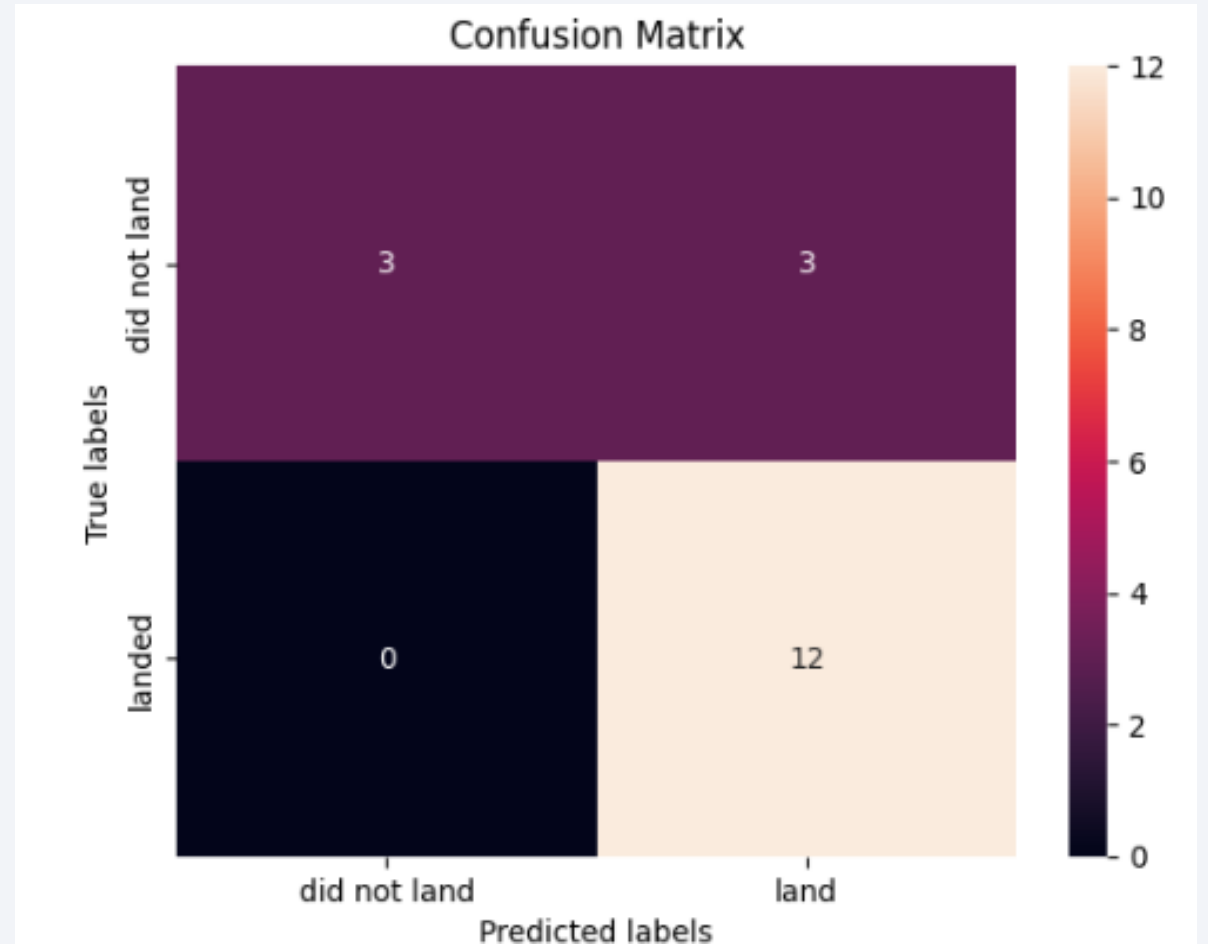


Decision Tree Classifier is overfitting since test accuracy is lower than train accuracy.

Accuracy values are too close in Logistic, SVM and K-NN to say which model is best. However for **K-NN**, the accuracy test and train are the closest indicating it generalizes better than the other two.

Confusion Matrix

- K-nearest neighbors classifier has the best Confusion Matrix with only three false positives.



Conclusions

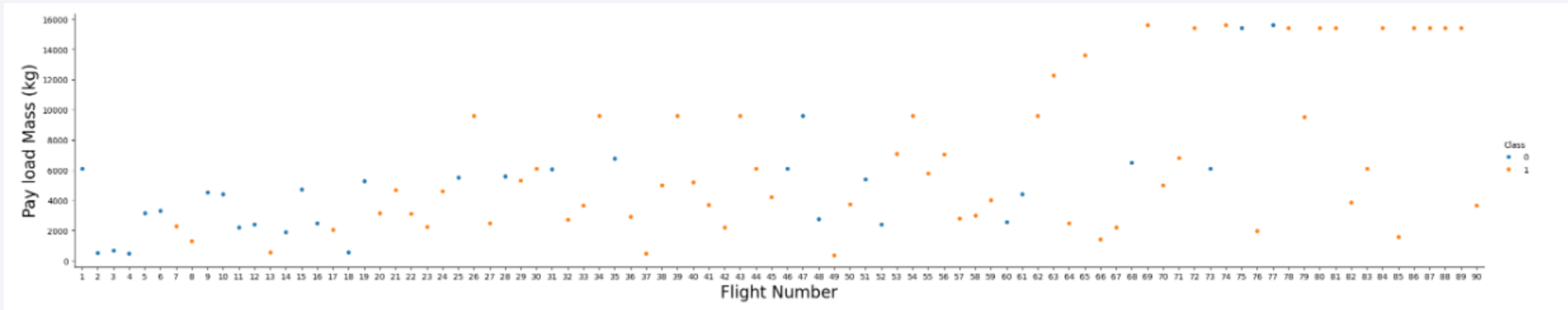
- The best tested model is K-nearest neighbors, with high accuracy of 84% and error are false positives and no false negatives. It generalized better than the other models.
- The payload with the highest launch success rate is from 2000 to 5000 kg.
- Booster FT has the highest success rate. Specification of that booster must be studied by the engineering team
- Most of the rockets with successful landings were launched from KSC that has 76.9% success rate. But it could be since it started working later. Further studies must be done to determine why it has greater success rate
- Launch site should be near the coast, min distances from different landmarks should be at least highway: 0.20 km, airport:22.56 km, city:19.29 km, railway:0.98 km
- Missions should be in orbits with greater success rate like ES-L1, GEO, HEO and SSO with 100% and avoid GTO orbit.

Appendix

In the appendix two graphs are included:

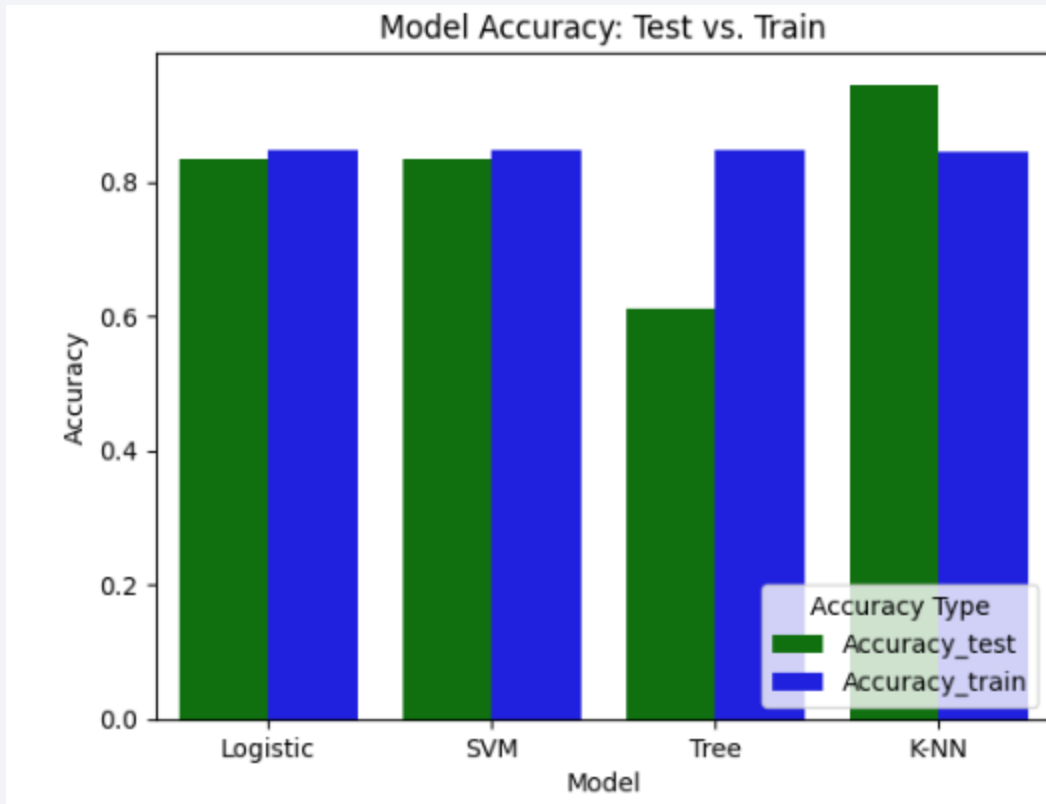
1. Pay load mass as function of Flight Number
2. Classification accuracy without stratify.

Pay load Mass (kg) vs. Flight Number



- As Flight Number Increases we see more Success landings (red dots) and pay load mass also increases. There is no relation between pay load mass and landing outcome.
- This will mean they learned from past failures (blue dots) and that they developed the capability to launch heavier payloads as time pass by. It could be because missions increased in difficulty, but is only a hypothesis.

Classification Accuracy without stratify



At first I thought from the bar plot that model K-NN was the best performed. However, if we compare train and test accuracy we understand that:

- Decision Tree classifier is overfitting,
- K-NN model has a strange behaviour it might be related to data leakage, so I will stratify the train test split.

Thank you!

